Investigation: FIBAM CIA INDUSTRIAL (CVM code 6017), fiscal year 2016, as it was on 2017-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Metalurgia e Siderurgia (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

Document used

2016 DFP, version 1, filed on 2017-03-27; origin: open data; type: individual; original scale: MIL.
Versions known on the date: 1 (2017-03-27).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=63592

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: component not computable: DEPI
- Alert by the study 1 rule: does not trigger alerts
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: —
- Percentile in group and year: —
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: low

DSRI, days' sales in receivables index
- Value: 1.079 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 87 (n = 19)
- Percentile in group and year: 66 (n = 425)
- The company's own history: median 0.948; Tukey range (1.5 × IQR): from 0.654 to 1.139; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

GMI, gross margin index
- Value: −0.131 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 3 (n = 19)
- Percentile in group and year: 3 (n = 438)
- The company's own history: median 1.099; Tukey range (1.5 × IQR): from 0.765 to 1.398; value below the range (index; 5 fiscal years)
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.214 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 76 (n = 19)
- Percentile in group and year: 84 (n = 426)
- The company's own history: median 1.043; Tukey range (1.5 × IQR): from 0.982 to 1.118; value above the range (index; 5 fiscal years)
- Confidence in the data: medium

SGI, sales growth index

Value: 0.873 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 39 (n = 19)
Percentile in group and year: 24 (n = 440)
The company's own history: median 0.891; Tukey range (1.5 × IQR): from 0.736 to 0.990; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

DEPI, depreciation index
Value: missing account: depreciation, cash flow lines in section 6.01.01
M-score component; it has no rule of its own
Percentile in sector and year: —
Percentile in group and year: —
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: low

SGAI, SG&A expenses index
Value: 1.986 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 97 (n = 19)
Percentile in group and year: 95 (n = 433)
The company's own history: median 1.051; Tukey range (1.5 × IQR): from 0.879 to 1.231; value above the range (index; 5 fiscal years)
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.469 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 3 (n = 19)
Percentile in group and year: 3 (n = 452)
The company's own history: median −0.067; Tukey range (1.5 × IQR): from −0.171 to 0.002; value below the range (index; 5 fiscal years)
Confidence in the data: medium

LVGI, leverage index
Value: 1.655 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 97 (n = 19)
Percentile in group and year: 97 (n = 446)
The company's own history: median 0.937; Tukey range (1.5 × IQR): from 0.633 to 1.271; value above the range (index; 5 fiscal years)
Confidence in the data: medium

DPO jump
Value: 30.8 days
Alert by the study 1 rule: yes
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 29.0 days)
Percentile in sector and year: 92 (n = 18)
Percentile in group and year: 91 (n = 410)
The company's own history: median 1.3 days; Tukey range (1.5 × IQR): from −25.3 days to 29.0 days; value above the range (5 fiscal years)
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 5.85 p.p.
Alert by the study 1 rule: no
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 61 (n = 19)
Percentile in group and year: 55 (n = 393)
The company's own history: median 6.73 p.p.; Tukey range (1.5 × IQR): from −0.49 p.p. to 17.25 p.p.; value inside the range (5 fiscal years)
Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.423 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 3 (n = 19)
Percentile in group and year: 3 (n = 449)
The company's own history: median −0.065; Tukey range (1.5 × IQR): from −0.169 to 0.002; value below the range (index; 5 fiscal years)
Confidence in the data: medium

Financial risk (context only)

Altman Z" (financial risk, context only)
Value: −7.658 (index)
Components: X1 = −0.123; X2 = −0.910; X3 = −0.526; X4 = −0.333 (indices)
Percentile in sector and year: 12 (n = 20)
The company's own history: median 0.639; Tukey range (1.5 × IQR): from −0.162 to 1.768; value below the range (index; 5 fiscal years)
- No cut-off zone: the available source does not define zones for Z"
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 2/9 (8 computable)
Signals:
- Return on assets (ROA): 0
- Operating cash flow on assets (CFO): 0
- Change in ROA (ΔROA): 0
- Cash above earnings (ACCRUAL): 1
- Change in leverage (ΔLEVER): 0
- Change in current ratio (ΔLIQUID): 0
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 0
- Change in asset turnover (ΔTURN): 1
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Restatement through the following year's comparative: no difference above the tolerance, or the

next year's DFP did not exist yet on the date.

What was known on the date

- DFP 2011, v1: 2012-03-14
- DFP 2011, v2: 2012-04-18
- DFP 2012, v1: 2013-03-12
- DFP 2013, v1: 2014-03-13
- DFP 2014, v1: 2015-03-25
- FCA 2016, v1: 2016-01-26
- DFP 2015, v1: 2016-03-29
- FCA 2016, v2: 2016-04-07
- FCA 2016, v3: 2016-05-18
- DFP 2016, v1: 2017-03-27

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

Auditor's report: Com Ressalva (qualified).
Excerpt checked literally: "Em nossa opinião, exceto pelos possíveis efeitos dos assuntos descritos no parágrafo "Base para opinião com ressalva", se houver, essas demonstrações do valor adicionado foram adequadamente elaboradas, em todos os aspectos relevantes, segundo os critérios definidos nesse Pronunciamento Técnico e são consistentes em relação às demonstrações financeiras tomadas em conjunto.Outras informações que acompanham as demonstrações financeiras e o relatório do auditorA administração da Companhia é responsável por essas outras informações que compreendem o Relatório da Administração."
Auditor in the 2014 FCA: Binah Sp Auditores Independentes.
Auditor in the 2015 FCA: Apice Auditores Independentes S/S.
Auditor in the 2016 FCA: JPPS Auditores Independentes S/S.
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2012: Padrão Auditoria S/S -> Narazzaqui & Cia. - Auditores. The outgoing firm audited 8 consecutive fiscal years (2004 to 2011), by the start and end dates in the FCA.
- 2013: Narazzaqui & Cia. - Auditores -> Binah Sp Auditores Independentes (transition: two firms in the 2013 FCA). The outgoing firm audited 1 consecutive fiscal years (2012 to 2012), by the start and end dates in the FCA.
- 2015: Binah Sp Auditores Independentes -> Apice Auditores Independentes S/S. The outgoing firm audited 2 consecutive fiscal years (2013 to 2014), by the start and end dates in the FCA.
- 2016: Apice Auditores Independentes S/S -> JPPS Auditores Independentes S/S. The outgoing firm audited 1 consecutive fiscal years (2015 to 2015), by the start and end dates in the FCA.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful

accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 36 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-10T19:35:59+00:00.

Today's information (outside the reference date)

Registry status: CANCELADA. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.